



# Power Users

The juice flows through all those components—don't neglect the source!

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So you've begun assembling your own PC—blazing fast processor, high-end video card, the works. All your dreams may go up in smoke, though—literally—if you don't power these devices right. Not many of us think too much about Power Supply Units (PSUs) when upgrading or building new PCs—but we should!

It's interesting to note that most devices will chug along at a slightly lower power than what they should be getting, but not for long. But what's worse is, a cheap power supply could be sending out too much voltage to your components, and this can result in dead components. So how should you go about buying a PSU, and what should you look for?

## The Terms

First things first: get used to the jargon you'll come across when shopping.

**MTBF:** Mean Time Between Failures is a measure of how stable a system is.

**Operating range:** The maximum and minimum input voltages within which the power supply works. The wider the operating range, the more stable it will be, and the less vulnerable to power surges.

**Load Regulation:** The change in output voltage when the output load varies from minimum to maximum, expressed in terms of nominal output voltage. A low load regulation means that the PSU delivers optimum voltage regardless of system configuration.

**Ripple:** The magnitude of AC voltage that is superimposed on the DC output voltage. PSUs that output low ripple are vital for systems with high-speed processors and memory chips.

**Hold-up time:** The time period during which output power supply remains within prescribed limits in the event of a power failure. The longer the hold-up time, the longer the PSU will power the system between the time that power fails and the UPS kicks in. This is pointless if you don't have a UPS, of course.

**PF and PFC:** A power supply that

has Power Factor Correction can efficiently convert current to power. The Power Factor is the ratio of true power (Watts) and apparent power (Volts x Amperes). Normal power supplies have a PF of around 0.7, which means there is always some wastage. A PF above 0.8 means there's a PFC, meaning higher efficiency.

**Rails:** PSUs supply voltages along the 3.3V, 5V, and 12V rails. The latter supply power to moving parts like fans and drive motors. Depending on the design, the motherboard draws power from all three rails in varying amounts. The 12V rail is the most important one, and many PSUs come with multiple 12V rails.

## Look For...

PSUs are rated in watts, with ratings in the range of 300 W for entry-level models to high-end 1.8 kW models. Don't be fooled by the wattage rating—a higher wattage does not imply better quality.

Branded PSUs, especially from Corsair, Cooler Master, and VIP are preferred. If you plan on using your system for regular surfing and some movie viewing, you don't need to be too choosy—entry-level PSUs should suffice.

If you're building a gaming rig, you need a reliable PSU for high-end GPUs, CPUs, RAM modules and the like. If you're building a multi-GPU system, look for SLI/CrossFire certified PSUs. As a rule of thumb, you should look at all the components you have, add up their power usages, and multiply the sum by 1.5 to give you some headroom.

You can use online power supply calculators to get an idea of your power requirements. These let you select your configuration and calculate the wattage rating of your PSU. A good example is [www.extreme.outervision.com/PSUEngine](http://www.extreme.outervision.com/PSUEngine). While doing your research, check out hardware-oriented sites like [www.xbitlabs.com](http://www.xbitlabs.com) and [www.tomshardware.com/us/](http://www.tomshardware.com/us/) to get a fair idea of power consumption of components at different loads. It is always better to overestimate the rating a little—that way your PSU won't be under too much strain, and will last longer.

## Also...

No matter which PSU you choose, be sure to check whether it is ATX (or the newer ATX 2.2) compliant. Also look at the connectors—your PSU should have sufficient 4-pin Molex connectors (for hard disks and optical drives) to future-proof your system. Some designs have modular connectors where the wires can be detached from the PSU—this makes for better flexibility and less clutter inside your cabinet. ■

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## Typical Component Power Usage

|                            |         |
|----------------------------|---------|
| NVIDIA GeForce 8800 GTX    | 131.5 W |
| NVIDIA GeForce 8800 GTS    | 105.7 W |
| NVIDIA GeForce 7900 GS     | 45.1 W  |
| NVIDIA GeForce 7600 GT     | 45.1 W  |
| ATI Radeon X1950 Pro       | 65.7 W  |
| ATI Radeon HD 2600 XT      | 48.6 W  |
| ATI Radeon X1650 XT        | 55.2 W  |
| Intel Core2 Extreme QX9650 | 89.8 W  |
| Intel Core2 Extreme QX6850 | 127.1 W |
| Intel Core 2 Duo E6700     | 61.7 W  |
| Intel Core2 Quad Q6600     | 87 W    |
| AMD Athlon 64 FX-62        | 130 W   |
| AMD Athlon 64 X2 4600+     | 69 W    |
| AMD Athlon 64 X2 3800+     | 31 W    |
| Motherboard                | 30 W    |
| DDR SDRAM                  | 10 W    |
| DDR2 SDRAM                 | 7.5 W   |
| Hard drive                 | 25 W    |
| CD drive                   | 20 W    |
| Combo drive                | 30 W    |